Analytical Chemistry Cumulative Exam

November 9, 2007

Please answer the following questions as succinctly as possible. Be sure to justify all assumptions. Clarity of thought and presentation will be an important factor in the grading of this exam. The Analytical Chemistry Features from 1 Jan through 1 June 2007 had several recurrent themes. These themes will be addressed in this exam.

1. (20 points) PDMS is a commonly used material for analytical devices.

a) What is PDMS?

As chemists, we must be familiar with molecular structure and function.
b) Draw the chemical structure for PDMS.

c) What are some advantageous properties that PDMS has for creation of microfluidic devices? d) What are some of the disadvantages? Give **one** strategy for overcoming the disadvantages noting that some disadvantages can be overcome with more than one strategy.

2. (20 points) In Wirth's discussion of new separation media, she mentions that electrochromatography can occur without a charged surface.

a) What is electrochromatography? Include in your answer the basis for analyte movement and separation.

b) If the surface is not charged, what potential detection difficulties could arise? Why do these arise?

3. (20 points) Later in Wirth's article she mentions a situation with a new device where *H* was sacrificed due to flow rates being too low.
a) What is *H* in this case?
b) *H* can be calculated by the well known Van Deemter equation. Write the exact van Deemter equation, define each symbol, and draw a graph illustrating why *H* was sacrificed due to low flow rate.

4. (20 points) Two articles describe mass spectrometric determination of proteins and peptides. Solution properties such as pH affect mass spectrometric results. In particular, it is useful to have a mental picture of the charge state above and below the pI for the peptide or protein. For the following dipeptide: Lysine•Glutamic Acid, draw all possible charge states. (You will be graded more on knowing the approximate structure and charge states than exact structures.) a) Circle the charge states that would be most prevalent at the pI for this dipeptide. b) Give the predominant charge states that would be observed in the mass spectrum for positive ion mode during ESI-MS at a pH of ~ 2 and pH of ~ 10.

5. (20 points) In the article about fluorinated compounds in the environment, a statement is made about spike/recovery experiments.
a) What is a spike and why is it used in analytical chemistry?

The article goes on to state that spikes should be performed at LOQ and 10*LOQ.
b) What is LOQ? Why not use LOD? What is the purpose for spiking near the LOQ? What information does it give you about the analytical method?